

Factor $x^2 + bx + c$

Connect the rectangle pattern to positive, negative and mixed-sign trinomials.

LEARNING TIME
45–65 minutes · work at your pace

YOU WILL NEED
Paper or a place to show your working

LEARNING GOAL	WHAT SUCCESS LOOKS LIKE
Factor monic trinomials and retain an outside common factor when needed.	I can justify the signed pair, explain a useful middle split and check by expansion.

Use both conditions—not just one

For a trinomial whose leading coefficient is 1, expansion gives a useful pattern.

$$(x + p)(x + q) = x^2 + (p + q)x + pq$$

To factor $x^2 + bx + c$, look for integers p and q that **multiply to c and add to b** . “Monic” means the leading coefficient is 1. This sum/product rule depends on that leading 1; a different leading coefficient needs the method in 3.6.

Write the pair you are trying, then check both conditions. You are not guessing a full answer blindly—you are narrowing a small list of factor pairs.

WORKED EXAMPLE

Build a positive factor pair

Factor $x^2 + 11x + 24$.

- 1 List factor pairs for the constant.

$$1 \cdot 24; 2 \cdot 12; 3 \cdot 8; 4 \cdot 6$$

The signed product must be 24. Since the middle term is positive, start with positive pairs.

- 2 Choose the pair with sum 11.

$$3 + 8 = 11$$

The other sums are 25, 14 and 10. Only 3 and 8 satisfy both conditions.

- 3 Write and expand the factors.

$$x^2 + 11x + 24 = (x + 3)(x + 8)$$

The middle products are $8x$ and $3x$; they combine to $11x$.

Make sign decisions before searching

Constant c	Middle coefficient b	Signs in the factors
Positive	Positive	Both positive
Positive	Negative	Both negative
Negative	Either	Opposite signs; the greater magnitude sets the sign of the sum.

For $x^2 - 9x + 20$, the product is positive and sum negative: use -4 and -5 . For $x^2 + x - 20$, use 5 and -4 . The same magnitudes can create different trinomials because the signs change the sum and product.

Factor a trinomial with opposite signs

1 Use opposite signs because the product is negative.

The larger magnitude must be negative to produce a negative middle coefficient.

2 Check the signed sum.

This matches the coefficient of x .

3 Write and verify the answer.

Expanding gives $x^2 + 5x - 7x - 35$, which simplifies to the original.

One legitimate method is to split the middle term, group the terms, and remove the shared binomial. For $x^2 + 7x + 12$, enter the full intermediate expression—not just two numbers.

Split the middle term into two terms that will allow grouping. Keep the complete expression.

Your answer

Type powers as x^2 and multiply with brackets: $(x+3)(x+4)$. For number questions, enter a number or product.

[illegible]

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Refresh a prerequisite, then return

The order of the two middle terms can change without changing the method. Another valid strategy can be kept on paper and checked using its final factors. You do not have to reproduce a preferred sequence simply to show that your mathematics works.

WORKED EXAMPLE

Do not stop after taking out the GCF

Factor $3x^2 + 21x + 36$ completely.

- 1** Remove the numerical GCF.

$$3x^2 + 21x + 36 = 3(x^2 + 7x + 12)$$

This is equivalent, but the trinomial inside can still be factored.

- 2** Factor the inside monic trinomial.

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

The pair multiplies to 12 and adds to 7.

- 3 Keep the outside factor.**

$$3x^2 + 21x + 36 = 3(x + 3)(x + 4)$$

Dropping the 3 would change every coefficient. Verify the entire product.

What went wrong?

Repair an answer

A student wrote $(x + 2)(x + 6)$. Repair the factorization of the polynomial below.

$$x^2+7x+12$$

Your answer

$$(x+3)(x+4)$$

Type powers as x^2 and multiply with brackets: $(x+3)(x+4)$. For number questions, enter a number or product.

$$(x+3)(x+4)$$

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Keep these ideas straight

Monic

A polynomial with leading coefficient 1.

Leading coefficient

The coefficient of the highest-degree term.

Decomposition

Splitting a term into equivalent parts to make grouping possible.

Equivalent expressions

Expressions with the same value for every allowed value of the variables.

Try it with less help each time

Use the support you need. You can also go directly to a fresh check; these practice steps do not lock your progress.

1. Finish the organised example

The constant product is 20 and the middle-term sum is 9. Begin a pair table: 1 and 20 give 21; 2 and 10 give 12. Continue the pattern and explain how the pair makes the two middle terms.

Complete the pair: the two integers add to 9 and multiply to 20.

$$x^2+9x+20$$

First integer

4

Second integer

5

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Write the complete expression with the middle term split for grouping.

$$x^2+9x+20$$

Your answer

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2. Set up the next one yourself

This time, choose the pair before opening the hint. Then write the factorization. Check it by expansion on paper.

Choose two integers that support factoring this trinomial.

$$x^2+10x+21$$

First integer

3

Second integer

7

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Now write the complete factorization.

$$x^2 + 10x + 21$$

Your answer

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[illegible]

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3. Use a genuinely different question

Select a fresh positive-monic check from the practice page. Previously seen or supported targets are excluded; no timer is required.

Open fresh-check choices

SHOW YOUR THINKING

Put it together

Try this without opening a hint first. Then explain one decision. You may use paper and keep that work for your teacher.

Lesson check

Factor completely over the integers.

$$x^2 - x - 12$$

Your answer

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Type powers as x^2 and multiply with brackets: $(x+3)(x+4)$. For number questions, enter a number or product.

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Last checked response: $(x-4)(x+3)$ · Correct in the requested form · without in-course mathematical support · unit3-algebra-2|unit3-contracts-2.1|bounded-factors-v1

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Explain how the signs of your factors produce both the middle term and the constant.

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 崑巾膠忘情憲挲掌慕敦晓杀桺櫚橢懶毡汴滄濺潯焰熿 犇獠琶産痠眈矧磚籽窰簪粹綰縉 𠂇耿脬蠹茆荃蔭藍蚰
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Short explanation · up to 160 characters

☐ I explained my method on paper instead.

Your lesson check has not been recorded.

Recording a check tracks work completed, not a grade or a claim of mastery. A response that needs repair is kept with that status.

Work recorded $\neq$ assignment submitted